FILTER DESIGN TEST CASE

FBAR (Film Bulk Acoustic Resonator) - Band 26 (850 MHz) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a FBAR (Film Bulk Acoustic Resonator) filter targeting Band 26 (850 MHz) for Automotive V2X Communication. The filter is designed on Quartz substrate with a target center frequency of 3996.0 MHz and bandwidth of 393.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: FBAR (Film Bulk Acoustic Resonator)
- Frequency Band: Band 26 (850 MHz)
- Application: Automotive V2X Communication
- Substrate: Quartz
- Package: QFN (Quad Flat No-Lead)
- Design Tool: Synopsys CustomSim
- Design Center: Irvine Design Center

This specification applies to all variants of the FBAR (Film Bulk Acoustic Resonator) design targeting Band 26 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 3996.0 MHz
Bandwidth (3 dB): 393.7 MHz
Insertion Loss Target: <= 1.3 dB
Return Loss Target: >= 18.0 dB
Out-of-Band Rejection: >= 48.0 dB
Isolation (Duplexer): >= 38.0 dB
Q Factor: 3285
Temperature Coefficient:-20.4 ppm/°C
Die Size: 2.1 x 1.4 mm
Wafer Size: 6-inch
Number of Resonators: 8
Electrode Material: Pt/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Synopsys CustomSim and load the FBAR (Film Bulk Acoustic Resonator) template project.
2. Define the substrate stack: Quartz with specified layer thicknesses.
3. Set design targets: center freq 3996.0 MHz, BW 393.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.3 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, Quartz).
10. Perform on-wafer probing using Keysight E5080B ENA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 11988 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.3 dB
- Return loss in passband shall be >= 18.0 dB
- Out-of-band rejection at +/- 591 MHz offset >= 48.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 10.2 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 94%

6. TEST RESULTS

Measurement Date: 2024-05-23
Devices Tested: 36
Insertion Loss (meas): 0.98 dB
Return Loss (meas): 19.6 dB
Rejection (meas): 50.5 dB
Isolation (meas): 43.5 dB
Q Factor (meas): 3285
Group Delay Variation: 4.7 ns
Power Handling: +28 dBm
Die Yield: 91.9%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

a) The FBAR (Film Bulk Acoustic Resonator) design demonstrated acceptable performance for Band 26 (850 MHz).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Group delay flatness improved compared to previous revision.
e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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